

EXPERIMENT 3

3

Classification of the Chemical Reactions as (i) Combination (ii) Decomposition (iii) Displacement (iv) Double Displacement Reaction**Aim**

To classify the given chemical reactions into the type (i) combination (ii) decomposition (iii) displacement and (iv) double displacement.

1. (A) Action of water on quick lime (CaO).
2. (B) Effect of heat on ferrous sulphate crystals.
3. (C) Reaction of copper sulphate solution with iron nails.
4. (D) Reaction between sodium sulphate and barium chloride solutions.

Requirements

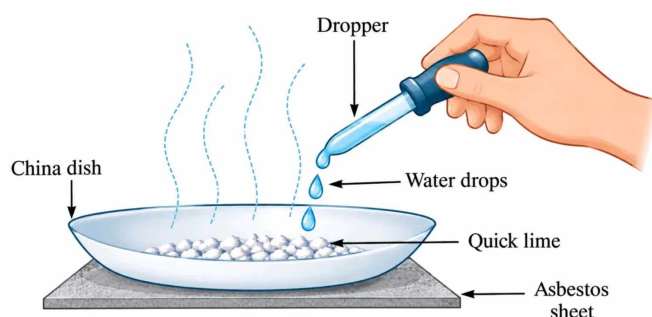
A 250 ml beaker, a china dish, an asbestos sheet, a dropper, hard glass test tubes, a test tube holder, a test tube stand, sandpaper, filter paper, and a burner or spirit lamp.

Chemicals Required

Quick lime, ferrous sulphate crystals, iron nails, copper sulphate solution, sodium sulphate solution, barium chloride solution, and water.

(A) Action of Water on Quick Lime**Procedure**

1. Place about 10 g of quick lime (CaO) into a clean china dish, and keep the china dish on an asbestos sheet.
2. Take some water in a beaker. Using a dropper, add a few drops of this water onto the quick lime in the dish and note down your observations.

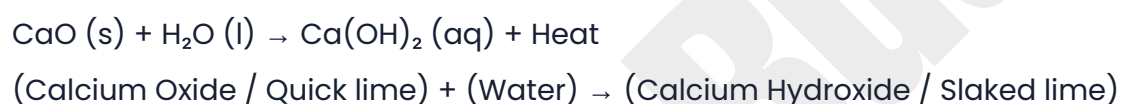
Diagram

Action of water on quick lime

Observation Table

Experimental Procedure	Observation
1. Water is added to quick lime.	Added water is converted into steam. China dish becomes hot. Hissing sound is produced.
2. The noise produced in the reaction.	Hissing sound.
3. The heat is absorbed or evolved during the reaction.	Evolved.
4. A gas or vapour is formed during the reaction.	Vapours are formed.
5. The change in the physical state.	Solid state gets converted into liquid state.

Chemical Reaction



Inferences

1. The reaction between quick lime and water is an exothermic reaction.
2. Here calcium oxide and water react to form calcium hydroxide (slaked lime).
3. This is an example of a combination reaction.
4. The clear solution obtained after the suspension settles is called lime water.

Precautions

1. Never put quick lime in water directly, as it may cause the lime water to splash out of the beaker and damage the eyes or skin.
2. Do not touch or dip your finger in the lime water.

(B) Effect of Heat on Ferrous Sulphate Crystals

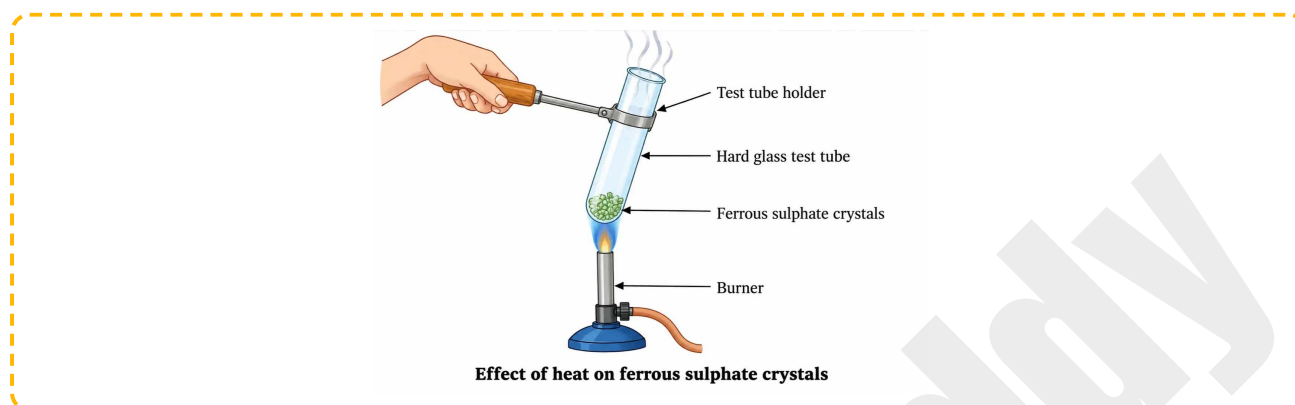
Procedure

1. Take about 10 g of ferrous sulphate crystals in a clean, dry hard glass test tube and note the colour of the crystals.

2. Hold the test tube with a test tube holder and heat it gently on a low flame for about 10 minutes.
3. Note the smell and colour of the gas that is evolved.
4. Keep heating until the colour of the residue inside the test tube changes.
5. Place the hot test tube on an asbestos sheet and observe the colour of the residue once it cools down.

Diagram

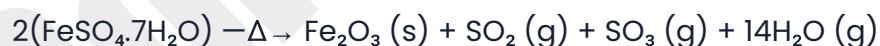
(Label the various components in the given diagram)



Observation Table

Experimental Procedure	Observation
1. The original colour of Ferrous sulphate.	Light green
2. Ferrous sulphate is heated.	White powder of anhydrous ferrous sulphate is formed. Finally it gives a mixture of Fe_2O_3 , SO_3 and SO_2 .
3. Colour of the gas evolved during heating.	Colourless
4. The colour of the substance in the cold test tube.	Light green

Chemical Reaction



Inferences

1. Light green crystals of ferrous sulphate undergo decomposition on heating. A mixture of SO_3 and SO_2 gases is formed.
2. A residue of red colour remains in the test tube.
3. This is an example of a decomposition reaction.

Precautions

1. Use a hard glass test tube for heating the substance.
2. Keep the test tube pointed away from yourself while heating.

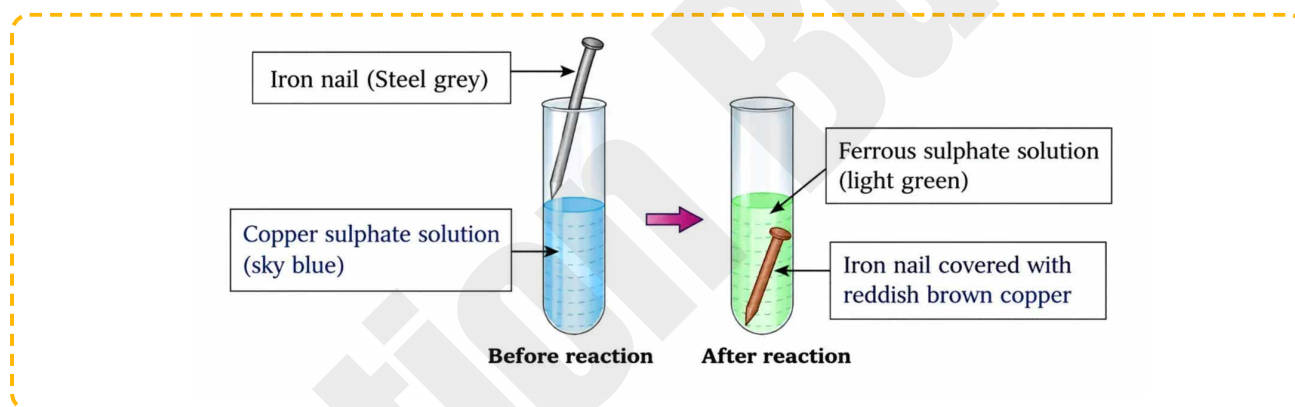
(C) Reaction of Copper Sulphate Solution on Iron Nails

Procedure

1. Take about 20 ml of copper sulphate solution in a test tube and note its colour.
2. Take 2 or 3 iron nails that are free from rust. Clean them using sandpaper and wash them with water, then note the colour of the iron nails.
3. Place the iron nails into the copper sulphate solution and leave them there for about 15 minutes.
4. Take the nails out of the solution, rinse them with water, and place them on a filter paper to dry.
5. Finally, note the colour of the solution as well as the colour of the nails after the reaction.

Diagram

(Label the various components in the given diagram - Before/After reaction)



Observation Table

Experimental Procedure	Observation
1. Colour of copper sulphate solution before the experiment.	The colour of copper sulphate solution before the experiment is sky blue.
2. Colour of iron nail before the experiment.	Colour of iron nail before the experiment is steel grey.
3. Colour of copper sulphate solution after the experiment.	Solution becomes light green after the experiment.

4. Colour of iron nail after the experiment.

The iron nail becomes reddish brown after the experiment.

Chemical Reaction



(Iron / steel grey) + (Copper sulphate / sky blue) → (Ferrous sulphate / light green) + (Copper)

Inferences

1. When the brown coloured iron nails are immersed in the blue coloured copper sulphate solution, they displace copper from the solution and their colour turns reddish.
2. This is a displacement reaction.

Precaution

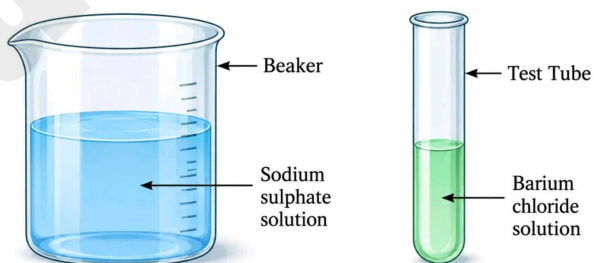
Clean the iron nails by rubbing them with sandpaper for an effective result.

(D) Reaction Between Sodium Sulphate Solution and Barium Chloride Solution

Procedure

1. Take about 15 ml of sodium sulphate solution in a clean beaker and note its colour and appearance.
2. Take about 5 to 10 ml of barium chloride solution in a clean test tube and note its colour and appearance.
3. Add the barium chloride solution to the sodium sulphate solution while stirring continuously. Observe and note the changes that take place.

Diagram

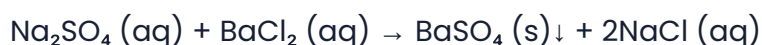


Reaction between sodium sulphate solution and barium chloride solution

Observation Table

Experimental Procedure	Observation
1. The colour and appearance of sodium sulphate solution before the reaction.	The colour and appearance is clear and transparent.
2. The colour and appearance of barium chloride solution before the reaction.	The colour and appearance is clear and transparent.
3. The colour and appearance of the mixture after mixing two solutions.	White precipitate.

Chemical Reaction



(Sodium sulphate / colourless) + (Barium chloride / colourless) → (Barium sulphate / white) + (Sodium chloride / colourless)

Inferences

1. In this chemical reaction, two new compounds are formed through the mutual exchange of the components (ions or radicals) of the two compounds. Such reactions are called double displacement reactions.
2. In this reaction, a white coloured insoluble compound, barium sulphate, is formed.
3. This is a double displacement reaction.

Precaution

Clean the apparatus thoroughly before using it.

Multiple Choice Questions

1. To observe the reaction of water on quick lime,

- (A) quick lime is added to water in a test tube
- (B) a lot of water is added to quick lime
- (C) a few drops of water are added to quick lime
- (D) none of these

Answer: (C) a few drops of water are added to quick lime

2. When crystals of Ferrous sulphate are strongly heated, the residue obtained is

- (A) red in colour
- (B) blue in colour
- (C) green in colour
- (D) colourless

Answer: (A) red in colour

3. When Barium chloride solution is added to Sodium sulphate solution,

- (A) blue precipitate is formed (B) white precipitate is formed
(C) green precipitate is formed (D) no reaction takes place

Answer: (B) white precipitate is formed

4. The reaction of iron nails with copper sulphate solution is an example of

- (A) combination reaction (B) decomposition reaction
(C) displacement reaction (D) double displacement reaction

Answer: (C) displacement reaction

5. Reddish brown deposit formed on iron nails kept in a solution of copper sulphate is

- (A) Cu_2O (B) Cu (C) CuO (D) CuS

Answer: (B) Cu

Remember

In double displacement reactions, two reactants exchange ions and form two new compounds.

Date: _____

Teacher's Signature: _____